DEN Data Visualization Color Palette

v1.0 Design Document

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1 Introduction

This document specifies the official DEN (Dewan Ekonomi Nasional) data visualization color palette, version 1.0. The palette provides 12 categorical colors organized in three tiers, two colormaps for continuous scales, and usage guidelines for economic charts.

The palette is implemented in three packages:

- **fig-den** (Python/matplotlib, v1.0)
- **ggden** (R/ggplot2, v1.0)
- **den-scheme** (Stata, v1.0)

All three packages use identical hex values in the same order.

2 Official DEN Color Identity

Singgih established the official DEN color identity with three colors in the gold-brown hue family (Figure 1).

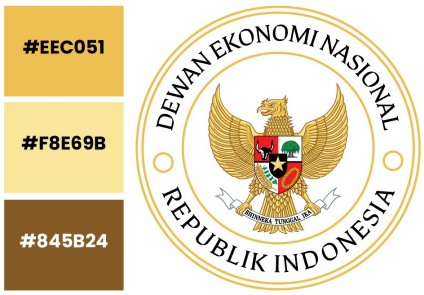

Figure 1: Official DEN color identity.

Table 1: The three official DEN brand colors.

Name	Hex	Description
Gold	`#EEC051`	Primary brand color
Light Gold	`#F8E69B`	Light/pale gold
Dark Brown	`#845B24`	Dark accent

These three colors work for logos and letterheads, but **data visualization requires colors that are immediately distinguishable** — especially in charts with 4–8+ categories. The gold-brown family alone cannot provide sufficient contrast for multi-series charts.

3 Design Principles

The palette follows a principle of **minimal departure from the brand**:

1. The three official colors anchor the palette.
2. Additional colors are derived from the gold-brown family (darkening, desaturation, or hue shifts).
3. One non-brown hue is introduced: red (`#C00000`), already used extensively in the DEN chartbook.
4. Every non-official color was validated against actual chartbook usage.

3.1 Color Extraction from the Chartbook

The Directorate for Acceleration of Priority Economic Programs produces a monthly economic chartbook. Colors were extracted from chart pages (pp. 5–28) of the March 2026 edition using PyMuPDF.

Table 2: Most frequently used colors in the March 2026 chartbook.

Hex	Count	Notes
`#FFC000`	259	Most-used bar fill in entire chartbook
`#935200`	125	Dark brown for bars/lines
`#855209`	85	Variant of above
`#C00000`	74	Red accent
`#F0A12E`	37	Close to old package gold
`#A4634E`	26	Close to old package brown
`#C17529`	25	Orange for tertiary series
`#91571F`	24	Dark amber variant

Key finding: the previous package gold (`#F0A22E`) was **not** the chartbook's primary gold (`#FFC000`). The official brand gold (`#EEC051`) sits between them. The v1.0 palette corrects this by using the official brand color as the primary and the chartbook workhorse as a secondary.

4 The 12-Color Palette

The palette has three tiers: simple charts (2–3 series) use the most brand-faithful colors, complex charts (8+ series) draw from the full set.

4.1 Tier 1: Core (positions 1–3)

For charts with 1–3 data series.

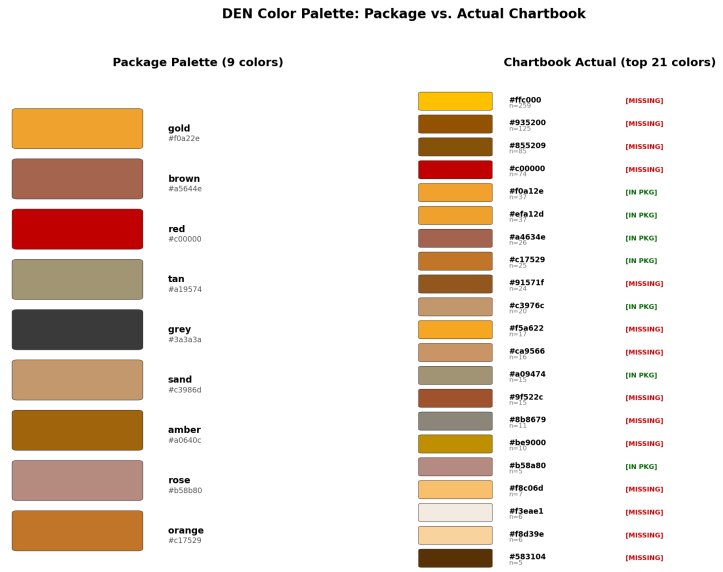

Figure 2: Old package palette (9 colors) vs. actual chartbook colors.

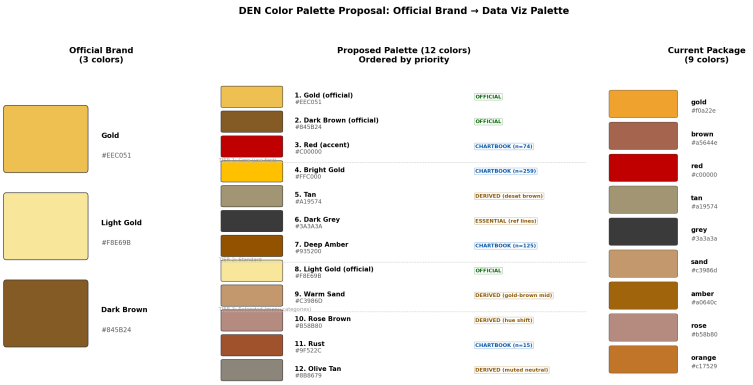

Figure 3: The DEN 12-color palette, compared with the official identity (3 colors) and old package (9 colors).

Table 3: Tier 1, core colors.

Position	Name	Hex	Source	Role
1	Gold	`#EEC051`	Official	Primary: first bar/line
2	Dark Brown	`#845B24`	Official	Secondary: second series
3	Red	`#C00000`	Chart-book	Accent: emphasis, negatives

Why Red at position 3? Light Gold (`#F8E69B`) is too pale for lines and small elements — it washes out on white backgrounds. Red provides the contrast needed for a usable 3-color set while remaining warm-toned. It is the chartbook's go-to accent (74 uses).

Figure 4 demonstrates the core 3-color set in a line chart.

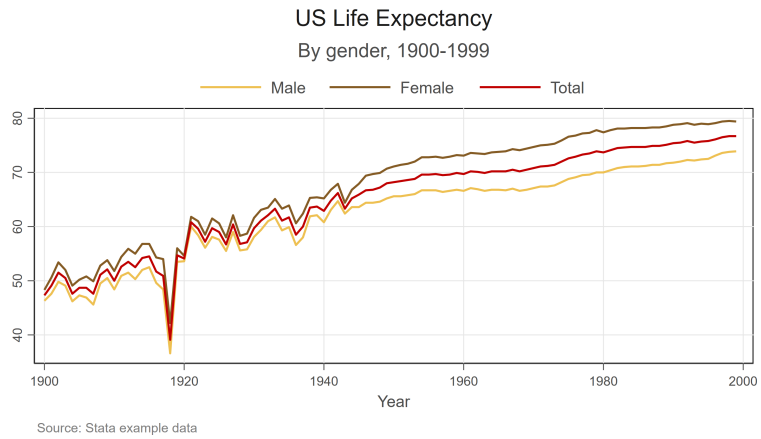

Figure 4: Line chart using core colors: Gold, Dark Brown, and Red.

4.2 Tier 2: Standard (positions 4–7)

For charts with 4–7 series.

Table 4: Tier 2, standard colors.

Position	Name	Hex	Source	Role
4	Bright Gold	`#FFC000`	Chart-book	High-saturation gold, bar workhorse
5	Tan	`#A19574`	Derived	Desaturated brown

Position	Name	Hex	Source	Role
6	Dark Grey	`#3A3A3A`	Essential	Reference lines, baselines, text
7	Deep Amber	`#935200`	Chart-book	Darker variant of official brown

Why Bright Gold (`#FFC000`) at position 4? This is the most-used color in the chartbook (259 occurrences). It is visually close to the official gold but more saturated, making it distinguishable in side-by-side comparisons.

Figure 5 and Figure 6 show bar charts using core and standard colors.

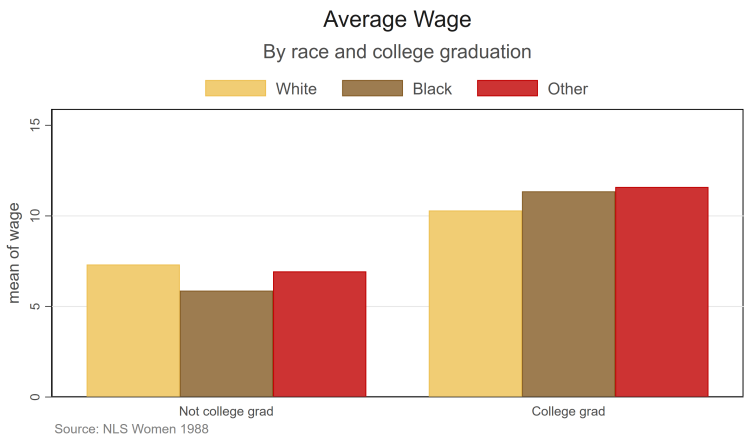

Figure 5: Grouped bar chart using 3 core colors.

4.3 Tier 3: Extended (positions 8–12)

For charts with many categories (trade partners, sectoral breakdowns, etc.).

Table 5: Tier 3, extended colors.

Position	Name	Hex	Source	Role
8	Light Gold	`#F8E69B`	Official	Area fills, lightest bar layer
9	Warm Sand	`#C3986D`	Derived	Midpoint gold-brown
10	Rose Brown	`#B58B80`	Derived	Hue-shifted for variety
11	Rust	`#9F522C`	Chart-book	Distinct warm accent
12	Olive Tan	`#8B8679`	Derived	Muted neutral

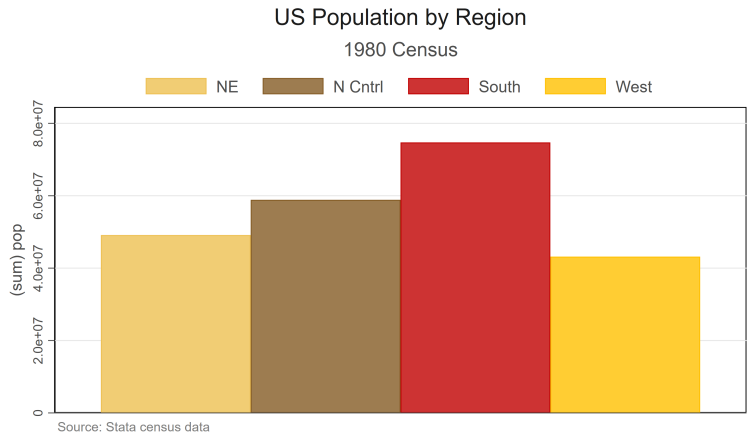

Figure 6: Four-category bar chart using core + standard colors.

Why Light Gold (`#F8E69B`) at position 8? As an official brand color it belongs in the palette, but its very light value makes it unsuitable for lines or narrow bars. At position 8, it is used only when many categories require it — typically as an area fill or the lightest segment in stacked bars, where its lightness is an asset.

Figure 7 shows all 12 colors in a multi-line chart.

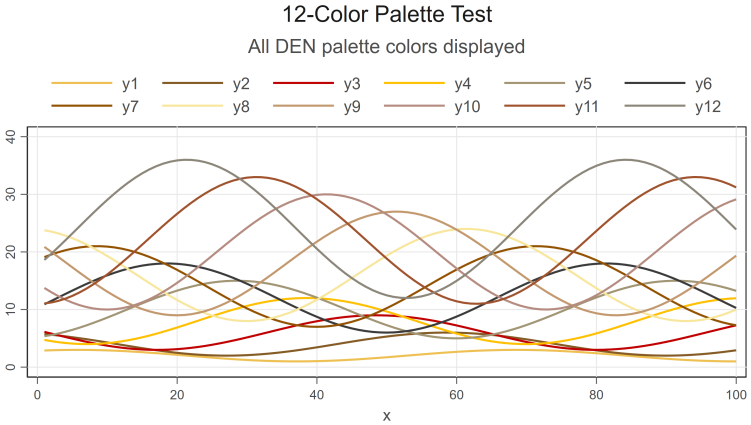

Figure 7: All 12 palette colors displayed in a line chart.

4.4 Colormaps / Gradients

For continuous scales (heatmaps, choropleths):

- **Sequential:** Light Gold (`#F8E69B`) → Gold (`#EEC051`) → Dark Brown (`#845B24`)
- **Diverging:** Gold (`#EEC051`) ← Light Gold (`#F8E69B`) → Red (`#C00000`)

Note on the diverging midpoint: Light Gold (`#F8E69B`) carries a warm visual bias. In economic charts where the midpoint has strict semantic meaning (e.g., zero growth, trade balance), consider forcing an off-white (`#F5F5F5`) at exactly zero for accurate visual encoding.

5 Design Evaluation

5.1 Strengths

The **core 3-color set (Gold, Dark Brown, Red) is excellent**. It provides immediate, unambiguous separation and covers the vast majority of DEN chart output. Gold anchors DEN identity, Dark Brown provides gravitas, and Red draws attention to highlights. This combination is particularly well-suited for policy charts (e.g., Indonesia vs. ASEAN vs. World).

The **brand identity is unmistakable**. The warm gold-brown palette is distinctly institutional and Indonesian — no one will confuse a DEN chart with a World Bank blue or IMF grey. The overall aesthetic reads as a bespoke publication rather than default software output.

For 2–4 series charts, the palette is very strong. Figure 5 and Figure 8 demonstrate clean, readable output with good contrast.

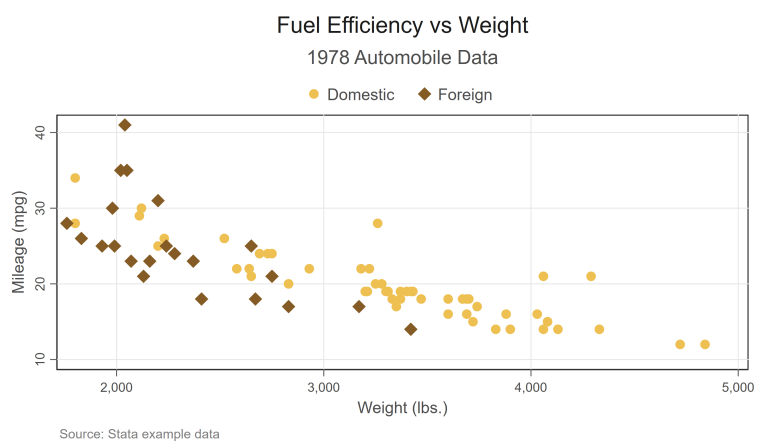

Figure 8: Scatter plot using 2 core colors with distinct marker shapes.

5.2 Hue diversity concern at 5+ series

Almost all 12 colors reside in a narrow slice of the color wheel (yellow → orange → red-brown). The palette contains **zero cool tones** — no blue, teal, or green. This creates two problems:

1. **Perceptual crowding.** At 5+ series, especially in line charts, the browns/tans/ambers merge into a muddy cluster. Positions 2 (Dark Brown), 5 (Tan), 7 (Deep Amber), 9 (Warm Sand), 11 (Rust), and 12 (Olive Tan) are all in the brown family. Figure 7 demonstrates that beyond 4 lines, individual series become difficult to identify by color alone.

2. **No semantic contrast pair.** Economic charts frequently need natural warm/cool contrast: exports vs. imports, surplus vs. deficit, growth vs. contraction. The current palette forces these distinctions to be made with "darker warm" vs. "lighter warm," which is perceptually weaker.

5.3 Gold saturation overlap

Positions 1 (Gold, `#EEC051`), 4 (Bright Gold, `#FFC000`), and 8 (Light Gold, `#F8E69B`) are three shades of yellow-gold. In Figure 6, the NE (pos 1) and West (pos 4) bars can read as "same category, different shade" at first glance. Three golds in a 12-color categorical palette is a luxury that reduces the effective perceptual range.

5.4 Light Gold visibility

Position 8 (`#F8E69B`) is too pale for lines and small markers on white backgrounds. It functions well as a fill color but is nearly invisible as a line, as visible in Figure 7 (y8).

5.5 Future direction

For a potential v2.0, consider replacing 1–2 colors in the Extended tier (positions 9–12) with muted cool tones. Candidates:

Current	Proposed replacement	Rationale
Warm Sand (pos 9)	Slate Blue (~`#4A6D7C`)	Cool contrast anchor
Olive Tan (pos 12)	Muted Teal (~`#5B8A72`)	Second hue family

This would give the palette 4 hue families (gold, brown, red, cool) instead of 2 (gold/brown, red) while keeping the DEN identity intact in the Core and Standard tiers. The warm institutional character comes from positions 1–7 — the Extended tier should prioritize readability.

6 Recommended Usage Patterns

Table 7: Recommended usage patterns.

Chart type	Colors to use
2-series	Gold + Dark Brown (positions 1–2)
3-series	Gold + Dark Brown + Red (positions 1–3)
Bar + line combo	Bars: Gold + Bright Gold, Line: Dark Brown
Stacked bar (4–6 categories)	Positions 1, 4, 2, 7 (light-dark alternation)
Many categories (8+)	Full 12-color palette

Best practice for 5+ series line charts: Prefer direct end-of-line labels over legend-only approaches. Because the palette is heavily warm-toned, colors in the brown family are easier to distinguish when labeled directly next to the data rather than matched to a distant legend.

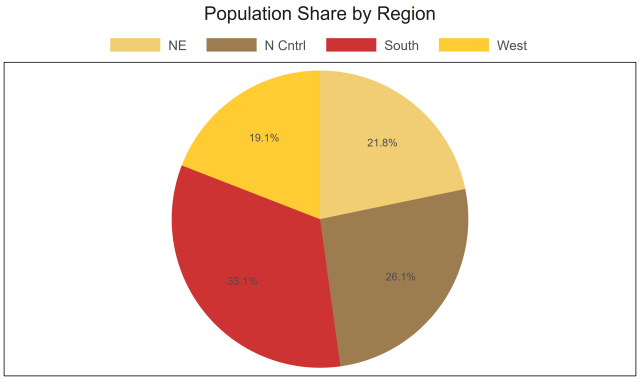

Figure 9: Pie chart using 4 colors from core and standard tiers.

7 Changes from v0.1

Table 8: Palette changes from v0.1 to v1.0.

Position	v0.1	v1.0	Change reason
1	`#F0A22E` gold	`#EEC051` gold	Aligned to official brand color
2	`#A5644E` brown	`#845B24` dark brown	Aligned to official brand color
3	`#C00000` red	`#C00000` red	Unchanged

Position	v0.1	v1.0	Change reason
4	`#A19574` tan	`#FFC000` bright gold	Chartbook's most-used color promoted
5	`#3A3A3A` grey	`#A19574` tan	Moved back; grey deferred to pos 6
6	`#C3986D` sand	`#3A3A3A` dark grey	Essential neutral kept
7	`#A0640C` amber	`#935200` deep amber	Replaced with chartbook-validated color
8	`#B58B80` rose	`#F8E69B` light gold	Official color added (was missing)
9	`#C17529` orange	`#C3986D` warm sand	Orange dropped (too close to golds)
10	—	`#B58B80` rose brown	Extended from 9 to 12 colors
11	—	`#9F522C` rust	Chartbook-validated addition
12	—	`#8B8679` olive tan	Muted neutral for large category sets

Dropped colors:

- `#A0640C` (old amber): too close to new Deep Amber and official Dark Brown.
- `#C17529` (old orange): too close to new Bright Gold and Gold.

8 Known Limitations

- **Hue clustering:** The palette is heavily warm-toned (yellow–brown–red). For charts with 5+ series, colors in the brown family may be difficult to distinguish. Use direct labeling and consider pattern fills to aid readability.
- **Colorblind accessibility:** The Gold/Bright Gold pair and the Brown/Amber pair may be hard to distinguish for viewers with deuteranopia. Use pattern fills or direct labels when accessibility is critical.
- **12-color ceiling:** Charts with 12+ categories will recycle colors. Group small categories into "Other" or use faceted panels.
- **Light Gold on white:** `#F8E69B` has low contrast on white backgrounds. Works for filled areas, not for lines or small markers.

9 Packages

Table 9: DEN visualization packages.

Package	Language	Version	Repository
fig-den	Python	v1.0	github.com/den-econ/fig-den
ggden	R	v1.0	github.com/den-econ/ggden
den-scheme	Stata	v1.0	github.com/den-econ/den-scheme

All packages use identical hex values in the same order.